

Report 6: Map

Nguyễn Thanh Hùng – 2540056

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1 Implementation

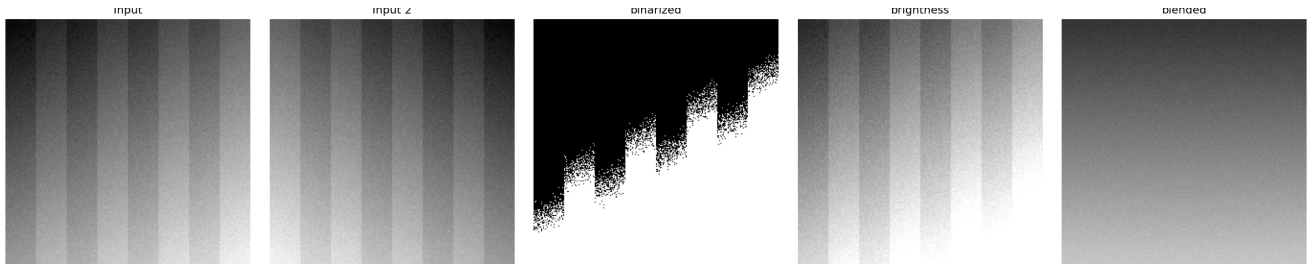
I used one thread per grayscale pixel. Binarization uses threshold 128. Brightness computes $1.2p + 30$ and clamps it to $[0, 255]$. Blending computes $0.5I_1 + 0.5I_2$; the second image is a horizontal flip of the first. I checked all outputs against NumPy.

I ran the notebook on Google Colab with a Tesla T4 and a 256×256 sample image. GPU timings use `time.time()` after warm-up and synchronization, averaged over three runs; allocation and image transfers are excluded. Where measured, CPU time is one run; speedup is relative to that Python CPU implementation.

2 Results

Maximum pixel error: 0 for all three operations.

Block	Binary (ms)	Brightness (ms)	Blend (ms)
(8, 8)	0.0629	0.0594	0.0936
(16, 16)	0.0466	0.0543	0.0584
(32, 16)	0.0405	0.0443	0.0550
(32, 32)	0.0649	0.0597	0.0685



3 Discussion

The 32×16 block gave the lowest time for all three operations. Blending was slower than binarization and brightness in this run; it also reads two input images.